

Technical Debt Plan

UrbanHeat Accra — Machine-Learning-Based Urban Heat Risk Prediction & Mitigation-Simulation Dashboard

Document Version: 1.0

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1. Introduction

This document outlines the technical debt accumulated during the development of the UrbanHeat Accra MVP, driven largely by the 48-hour development constraint. It classifies the debt and provides a roadmap for repayment in future iterations of the software.

2. Technical Debt Register

Debt ID	Technical Debt	Cause	Impact	Priority	Classification	Proposed Resolution	Target Version
TD-01	Synthetic training data instead of real satellite-derived data	48-hour constraint; no access to curated dataset	Model predictions are illustrative, not calibrated to real-world conditions	High	Scheduled for Future Resolution	Integrate real Landsat/Sentinel satellite-derived features and ground-truth temperature measurements	v2
TD-02	Global feature importance instead of per-instance SHAP values	Development time constraint; SHAP library adds complexity	Explanation is the same for all locations rather than location-specific	Medium	Acceptable Temporarily	Integrate SHAP library for local interpretable explanations	v1
TD-03	No authentication or authorisation	FR9 descope; MVP is read-only public tool	No access control; all endpoints publicly accessible	Medium	Scheduled for Future Resolution	Implement JWT authentication with role-based access control	v2

TD-04	Simplified linear NDVI simulation model	Physical simulation requires domain expertise beyond scope	Simulation results are indicative, not physically accurate	Medium	Acceptable Temporarily	Partner with environmental scientists for physics-based vegetation-temperature models	v2
TD-05	No rate limiting on API endpoints	Development time constraint	Potential for API abuse or denial of service	Low	Scheduled for Future Resolution	Add FastAPI rate limiting middleware (e.g., slowapi)	v1
TD-06	SQLite in production (Render)	Zero-config database for rapid deployment	Not suitable for concurrent writes; file-based, no connection pooling	Medium	Scheduled for Future Resolution	Migrate to PostgreSQL on Render	v1
TD-07	No automated CI/CD pipeline	48-hour constraint; manual deployment	Risk of deployment inconsistencies; no automated test enforcement	Low	Scheduled for Future Resolution	Set up GitHub Actions for automated testing and deployment	v1
TD-08	Limited automated test coverage	Time constraint; focused on critical API paths	8 test cases cover core paths but not edge cases or frontend	Medium	Scheduled for Future Resolution	Expand test suite; add frontend component tests with Vitest	v1
TD-09	No real-time data ingestion	Live satellite data integration is complex and out of scope	Data is static; no automatic updates when conditions change	High	Scheduled for Future Resolution	Build a data pipeline for periodic satellite data refresh	v2
TD-10	Single-city deployment	MVP scope limited to Accra	System cannot be used for other cities without manual data preparation	Low	Scheduled for Future Resolution	Generalise data model and UI for multi-city support	Lo te
TD-11	No content security policy (CSP) headers	Not configured	Minor XSS vulnerability surface	Low	Scheduled for Future Resolution	Add security headers middleware	v1

TD-12	Frontend mobile responsiveness incomplete	Time constraint; optimised for desktop first	Some components may not display optimally on very small screens	Low	Acceptable Temporarily	Complete responsive implementation with mobile-specific patterns	v1
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3. Technical Debt Repayment Roadmap

Immediate Action Items

- Run the API test suite and document baseline functional status.
- Verify environment variables and CORS configuration in Render/Vercel.

Version 1.1 (Short-Term Focus: Stability and Security)

- **TD-02:** Switch from global `feature_importances_` to SHAP for accurate local explanations.
- **TD-05, TD-11:** Implement API rate limiting and security headers to harden the public endpoints.
- **TD-06:** Provision a managed PostgreSQL instance and update `DATABASE_URL`.
- **TD-07, TD-08:** Establish GitHub Actions CI/CD and expand test coverage.
- **TD-12:** Refine mobile CSS (bottom sheets, card lists instead of tables).

Version 2.0 (Medium-Term Focus: Data Fidelity and Features)

- **TD-01, TD-09:** Build a data pipeline to ingest real Landsat-8/Sentinel-2 imagery, replacing the synthetic dataset.
- **TD-03:** Introduce an Admin role with JWT authentication to manage location data.
- **TD-04:** Replace linear simulation with a robust physics-based thermal model.

Long-Term Vision

- **TD-10:** Abstract the data schema to support multiple cities and regions dynamically.
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End of Technical Debt Plan